

COURSE TITLE : SUSTAINABLE ARCHITECTURE
COURSE CODE : 5184
COURSE CATEGORY : E
PERIODS/WEEK : 4
PERIODS/SEMESTER : 52
CREDITS : 4

TIME SCHEDULE

Module	Topics	Period
1	Concepts of sustainability	13
2	Sustainable planning & Design	13
3	Sustainable Building Materials and Construction	13
4	Recycling and Reuse	13
TOTAL		52

Course Objective :

Sl.	Sub	Student will be able to
1	1	Understand the concept of sustainability
	2	Appreciate the various green rating systems for sustainability
	3	Understand the properties, Uses and Examples of Sustainable Materials
2	1	Categorise different stages of construction process
	2	Comprehend the application of sustainability at different stages
	3	Appreciate the conservation of energy resources
3	1	Know the importance of waste reuse
	2	Conduct Sustainable Development Case Studies

Upon completing subject, the students should be able to

MODULE 1

1.1.0. Understand the concept of sustainability

- 1.1.1. Know the global environment and effects of climate change
- 1.1.2. Study the types of energy systems

MODULE II

2.1.0. Appreciate the various green rating systems for sustainability

- 2.1.1. Understand the importance of Sustainable Development.
- 2.1.2 Discuss the Sustainable approach to site planning and design
- 2.1.3 Understand the Model ecosystem of the site
- 2.1.4 Evaluate the environmental monitoring and testing during construction
- 2.1.5 Identify the Design facility within social and environmental thresholds

MODULE III

3.1.0 Understand the properties, Uses and Examples of Sustainable Materials

- 3.1.1 Understand Roles of Principles to improve the energy efficiency
- 3.1.2 Adopt the techniques of sustainable construction
- 3.1.3 Understand the alternative materials and construction methods

3.2.0 Discuss the use of local materials

3.3.0. Comprehend the application of sustainability at different stages

MODULE IV

4.1.0 Categorise different stages of construction process

- 4.1.1 Understand the Pre building, Building, Post building stages
- 4.1.2 Comprehend the application of sustainability at different stages

4.2.0 Understand the Architectural Reuse

- 4.2.1 Discuss the Construction and Demolition recycling

4.3.0 Appreciate the conservation of energy resources

- 4.3.1 Conservation of natural and building resources

4.4.0 Know the importance of waste reuse

- 4.3.1 Elimination of waste and minimize pollution
- 4.3.2 Identify the Innovative reuse of various wastes

4.4.0 Conduct Sustainable Development Case Studies

CONTENT DETAILS

MODULE -I

Concepts of sustainability : Energy and Global environment - Energy use and Climate change - Its impact - Types of Energy systems - Concept of Sustainability - Principles of conservation -synergy with nature - Bioregionalism - community basis shelter technology within bioregional patterns and scales – Ethical & environmental degradation - Green architecture and various international rating systems for sustainability- BREAM(UK) - CASBEE (Japan) - LEED (US) - Indian systems – TERI GRIHA rating – IGBC LEED India rating.

MODULE -II

Sustainable planning & Design : Sustainable Development -Sustainable approach to site planning and design - site inventories- relationships between site factors – development impacts from one area of the site on the other areas - Model ecosystem of the site -environmental monitoring and testing during construction- phasing of development -limits of change - Design facility within social and environmental thresholds

MODULE -III

Sustainable Building Materials and Construction : Properties - Uses and Examples of Primary, secondary and Tertiary Sustainable Materials - Principles to improve the energy -efficiency - siting and vernacular design, shade, ventilation, earth shelter, thermal inertia and air lock entrances - Techniques of sustainable construction – technologies - methods of effectiveness, and design synthesis – alternative materials and construction -methods: solar water heating panels; photovoltaic electricity generation; use of local - materials and on site growth of food, fuel and building materials.

MODULE -IV

Recycling and Reuse : Post building stages - Architectural Reuse - Waste prevention - Construction and Demolition recycling- Conservation of natural and building resources -Energy and material savings - types of wastes - Elimination of waste and minimize pollution- various Decomposing methods – Innovative reuse of various wastes

Case Studies: Sustainable Development Case Studies - illustrated examples of the planning, development and construction.

REFERENCES:

1. B.C.Bose, “Integrated approach to sustainable Development”, Rajat Publications, Delhi
2. Laurie Baker’s, “Chamoli Earthquake hand book”, Costford, Centre of science and technology for rural development.
3. Fuller Moore, “Environmental control systems Heating, Cooling, Lighting”. McGraw Hill, Newyork.
4. Caring A.Langston, Grace K.C.Ding, “Sustainable practices in built environment